

A Computational and Conformal Analysis of Multibrot Sets

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1 Abstract

The Mandelbrot set is the connectedness locus of the quadratic family $z \mapsto z^2 + c$, equivalently the set of parameters $c \in \mathbb{C}$ for which the critical orbit of 0 remains bounded. Its higher-degree analogue, the multibrot set M_j associated with $z \mapsto z^j + c$, has long served as a natural testbed for polynomial dynamics. These sets are known to be invariant under complex conjugation, to possess $(j - 1)$ -fold rotational symmetry, and to converge geometrically to the closed unit disk as $j \rightarrow \infty$. In this paper, I investigate the geometry of M_j from computational, geometric and analytic perspectives. Using large-scale numerical experiments, I study the growth of multibrot area as a function of degree and compare the results with the known disk-like limiting behavior. I also give explicit elementary derivations of reflectional and rotational symmetries for representative degrees and visually demonstrate the preservation of polynomial structure through equivalent surface maps. I then employ a conformal-analytic framework based on Laurent series and Gronwall’s area formula to study the geometry of M_j . By computing Laurent coefficients of the associated conformal map, I obtain independent upper bounds on multibrot areas and show that the proposed empirical area curve remains within these bounds across the computed degrees. Together, these computational, geometric, and analytic results provide a unified perspective on the area, symmetry, and large-degree geometry of multibrot sets.

2 Introduction

The Mandelbrot set plays a central role in complex dynamics and fractal geometry. For the quadratic family

$$f_c(z) = z^2 + c,$$

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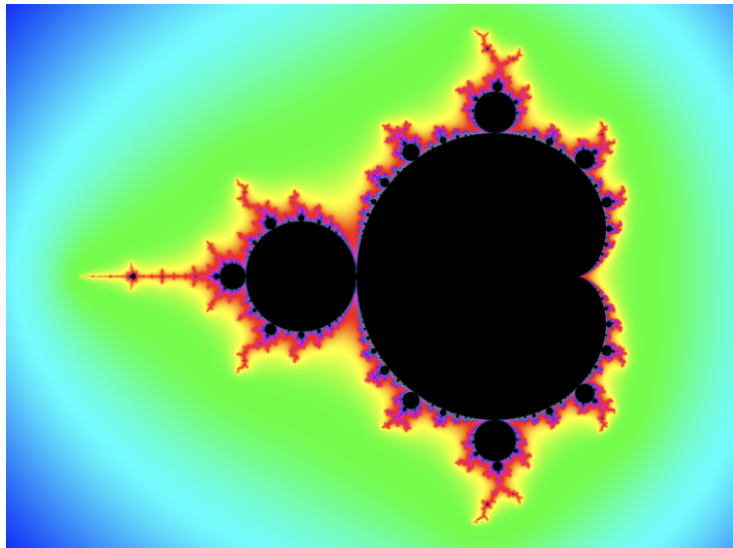


Figure 1: A high-resolution rendering of the classical Mandelbrot set ($j = 2$), highlighting its aesthetic beauty. (Source: Geek3 via Wikimedia Commons [11], licensed under CC BY-3.0.)

it may be defined as the set of parameters $c \in \mathbb{C}$ for which the critical orbit of 0 remains bounded under iteration. Equivalently, it is the connectedness locus for the family of quadratic polynomials, a point of view developed in the foundational work of Douady and Hubbard [9, 7, 8]. Beyond its visual appeal (shown in Figure 1), the Mandelbrot set encodes deep information about the global organization of polynomial dynamics and has motivated a large body of work on Julia sets, parameter spaces, bifurcation, and conformal geometry [14, 16, 4].

A natural generalization is obtained by replacing the quadratic family with the unicritical family

$$f_{c,j}(z) = z^j + c, \quad j \geq 2.$$

Its connectedness locus is the *multibrot set*

$$M_j = \{c \in \mathbb{C} : \text{the orbit of } 0 \text{ under } f_{c,j} \text{ is bounded}\}.$$

These higher-degree analogues have long been recognized as fundamental objects in holomorphic dynamics. In particular, they retain several structural features of the classical Mandelbrot set, including reflection symmetry with respect to the real axis and $(j-1)$ -fold rotational symmetry, while their geometry becomes increasingly disk-like as $j \rightarrow \infty$ [16, 3]. At the same time, many quantitative questions about their geometry remain much less explicit than in the quadratic case. Even for the classical Mandelbrot set, for example, no closed-form expression for its area is known; the most precise results rely on analytic expansions and numerical computation [10, 1, 19].

This paper studies the geometry of M_j from complementary computational and conformal perspectives. Computationally, it investigates the evolution of multibrot area and symmetry as the polynomial degree increases. Conformally, it employs Laurent-series methods to independently validate the numerical area computations through classical techniques from complex analysis. First, I use the escape-time algorithm to generate multibrot sets for degrees $j = 3, \dots, 20$, and I estimate their areas by systematically refining grid resolution and iteration depth until the numerical values stabilize. The resulting estimates increase monotonically from approximately 1.8 at $j = 3$ toward values close to π , consistent with the geometric limit $M_j \rightarrow \mathbb{D}$ as $j \rightarrow \infty$ [3]. A fit of the form

$$C(j) = \pi - \frac{\beta}{j^\alpha}$$

matches this trend well over the computed range, and additional simulations at $j = 50$, $j = 80$ and $j = 200$ remain consistent with this model.

Second, I analyze the symmetry of M_j by direct algebraic methods. Although reflectional and rotational invariance are classical facts in the theory of multibrot sets, the arguments are often left implicit or treated as immediate consequences of the defining family. Here I record these symmetries explicitly and derive them in a self-contained way from the iteration formula itself. I further identify regular $(j - 1)$ -gon patterns associated with the geometry of M_j , giving a concrete manifestation of the rotational symmetry. As j increases, these preferred directions become increasingly dense, offering an elementary geometric explanation for the observed approach to the unit disk.

Finally, I investigate multibrot sets from a conformal perspective. Building upon the Laurent-coefficient framework developed for the classical Mandelbrot set and its subsequent generalization to multibrot sets, I adapt an existing implementation to arbitrary polynomial degrees and use it to reconstruct multibrot boundaries and obtain Gronwall-based area bounds. Rather than studying the Laurent coefficients themselves, this framework is employed as an independent conformal-analytic validation of the numerical area computations developed throughout the paper.

Specifically, this paper makes the following contributions:

1. **Area computations across degree.** I compute numerical approximations of the area of M_j for increasing values of j and give empirical evidence for convergence toward π , consistent with the known disk-like limit [12, 3].
2. **Numerical Convergence Analysis** I investigate the convergence of escape-time area estimates by analyzing the effect of increasing the iteration limit, providing empirical evidence of numerical stabilization and relating these observations to the theoretical convergence of the finite-iteration approximations.
3. **Direct symmetry derivations.** For the cases $j = 4 - 6$, I give explicit algebraic derivations of reflectional and rotational symmetries and

show how these invariances propagate through iteration to realize the full dihedral group D_{j-1} .

4. **A geometric explanation for circular convergence.** I connect the growth in symmetry directions with the increasingly circular shape of M_j , providing a simple geometric interpretation of its large-degree behavior [3].
5. **Conformal-analytic validation through Laurent coefficients.** I adapt a quadratic Laurent-coefficient algorithm [20] to arbitrary Multibrot degrees, verify the adaptation through the inverse Böttcher function [5] and boundary reconstruction, and apply Gronwall's Area Theorem [10, 2] to establish independent analytic upper bounds that validate the numerical estimates.

The paper is organized as follows. Section 4 introduces the necessary definitions and background on iteration, boundedness, and escape criteria. Section 5 presents the escape-time algorithm and the numerical procedure used to estimate area. Section 6 discusses the resulting area data and its asymptotic behavior. Section 7 establishes the reflectional and rotational symmetries of M_j and derives the associated geometric patterns. Section 8 records several further geometric features suggested by the computations and develops a geometric explanation for the observed convergence toward circularity. Section 9 investigates multibrot sets through conformal geometry, Laurent series, and the inverse Böttcher map, culminating in an independent validation of the area computations through Gronwall's area formula.

3 Related Work

The Mandelbrot set and its higher-degree analogues, the multibrot sets, are central objects in complex dynamics. Foundational work of Douady and Hubbard established the modern framework for polynomial-like dynamics and connectedness loci, placing the Mandelbrot set at the center of the theory of complex polynomial iteration [9, 7, 8]. Subsequent work developed the structure of Mandelbrot and multibrot parameter spaces in much greater depth, including questions of local connectivity, fibers, and rigidity for unicritical polynomials [16, 6, 14].

For multibrot sets in particular, one of the most relevant asymptotic results is due to Boyd and Schulz, who proved that as $j \rightarrow \infty$, the multibrot sets converge geometrically to the closed unit disk, while the associated Julia sets converge to the unit circle in the Hausdorff metric [3]. This gives a precise description of the large-degree regime and confirms the heuristic picture that higher-degree multibrot sets become increasingly disk-like. Expository discussions emphasize the same visual phenomenon and highlight the increasingly circular appearance of the boundary as the degree grows [12].

A related question concerns quantitative geometric properties, especially area. For the classical Mandelbrot set, Ewing and Schober gave a rigorous

computational treatment of area using coefficients of the uniformizing map at infinity [10], and later work refined the numerical and analytic understanding of this problem [1, 19]. Even in the quadratic case, however, no closed-form expression for the area is known. By comparison, similar quantitative questions for multibrot sets appear to have received much less attention, particularly in the large-degree setting.

Symmetry is another basic feature of multibrot geometry. Reflection across the real axis and $(j - 1)$ -fold rotational invariance are standard properties of these sets, and they also appear in expository and computational discussions [12, 15]. What seems less emphasized in the literature is an elementary treatment of how these symmetries appear directly from the iteration formula and how they relate to visible geometric patterns in the parameter plane. This is the point of view taken in the present paper.

A complementary line of research studies the Mandelbrot set through conformal geometry. Jungreis established the exterior Riemann map of the complement of the classical Mandelbrot set, providing the Laurent-series framework associated with this conformal mapping [13]. Building on this framework, Ewing and Schober used Laurent coefficients together with Gronwall’s area formula to derive rigorous upper bounds for the area of the classical Mandelbrot set and compared these estimates with numerical pixel-counting methods [10]. More recently, Shimauchi generalized the Laurent-coefficient framework to arbitrary Multibrot sets, developing the corresponding coefficient theory and investigating arithmetic properties of the coefficients [17].

Building upon these works, the present paper employs the Laurent-coefficient framework for a different purpose. Rather than studying the coefficients themselves, they are used to reconstruct Multibrot boundaries and obtain Gronwall-based area bounds that provide an independent conformal-analytic validation of the numerical escape-time area computations across polynomial degree.

4 Definitions

Definition 1 *Let $j \geq 2$ be an integer and let*

$$f_{c,j}(z) = z^j + c, \quad c \in \mathbb{C}.$$

Starting from $z_0 = 0$, define the orbit recursively by

$$z_{n+1} = z_n^j + c \quad (n \geq 0).$$

The sequence $(z_n)_{n \geq 0}$ is called the orbit of 0 under $f_{c,j}$.

Definition 2 *The j th generalized Mandelbrot set, or multibrot set, is*

$$M_j = \{ c \in \mathbb{C} : (z_n)_{n \geq 0} \text{ remains bounded} \},$$

where (z_n) is the orbit defined by

$$z_0 = 0, \quad z_{n+1} = z_n^j + c.$$

Definition 3 A sequence $(z_n) \subseteq \mathbb{C}$ is called *bounded* if there exists a real number $R > 0$ such that

$$|z_n| \leq R \quad \text{for all } n \geq 0.$$

To make this more concrete, consider the quadratic case $f_c(z) = z^2 + c$. Starting from $z_0 = 0$, the first few iterates are

$$\begin{aligned} z_1 &= c, \\ z_2 &= c^2 + c, \\ z_3 &= (c^2 + c)^2 + c, \\ z_4 &= ((c^2 + c)^2 + c)^2 + c. \end{aligned}$$

Even after just a few steps, these expressions become quite complicated. For this reason, it is more practical to study whether the orbit stays bounded by using an escape criterion. In the next section, I prove that if an iterate ever satisfies $|z_n| > 2$, then the orbit escapes to infinity. This shows that every multibrot set M_j is contained in the closed disk of radius 2.

5 Escape Criterion and Computation

5.1 Escape Criterion

To approximate multibrot sets numerically, we need a practical criterion for deciding when a point is not in the set. The key fact is that once an iterate leaves the disk of radius 2, the orbit must diverge.

Lemma 5.1 *For every $j \geq 2$, the multibrot set M_j is contained in the closed disk*

$$\{c \in \mathbb{C} : |c| \leq 2\}.$$

Moreover, if for some n we have $|z_n| > 2$, then the orbit (z_n) diverges to infinity.

First observe that $z_1 = c$. Therefore, if $|c| > 2$, then already $|z_1| > 2$, so it is enough to show that any orbit with an iterate outside the disk of radius 2 escapes to infinity.

Assume that $|z_n| > 2$. Since $z_{n+1} = z_n^j + c$, the reverse triangle inequality gives

$$|z_{n+1}| = |z_n^j + c| \geq |z_n|^j - |c|.$$

Now if $c \in M_j$, then necessarily $|c| \leq 2$, so

$$|z_{n+1}| \geq |z_n|^j - 2.$$

Because $|z_n| > 2$ and $j \geq 2$, we have

$$|z_n|^j - 2 > |z_n|.$$

Hence $|z_{n+1}| > |z_n|$. Repeating this argument shows that the moduli continue to increase, so the orbit diverges to infinity.

Therefore, any parameter c for which some iterate satisfies $|z_n| > 2$ does not belong to M_j , and thus $M_j \subseteq \{c : |c| \leq 2\}$.

5.2 The Escape Time Algorithm

The escape criterion gives a practical way to approximate multibrot sets numerically. Although it is not possible to check directly whether an orbit remains bounded for infinitely many iterates, we can determine that a point does not belong to M_j as soon as one iterate leaves the disk of radius 2. This idea leads to the escape-time algorithm, which is the main computational method used in this paper.

My implementation follows the standard escape-time approach and is written in Python using NumPy for vectorized computation. The procedure is as follows:

1. Generate a grid of complex numbers representing a region of the complex plane.
2. Set $z_0 = 0$ at each grid point and iterate the map

$$z_{n+1} = z_n^j + c.$$

3. For each point c , check whether the orbit escapes the disk of radius 2 within the prescribed number of iterations.

Points whose orbits do not escape within the iteration limit are treated as belonging to the multibrot set for the purposes of visualization and numerical approximation. The code used to generate these sets is shown in Figure 2.

```

+ Code + Text
[1] import matplotlib.pyplot as plt
import numpy as np
import warnings

warnings.filterwarnings("ignore")

[2] def complex_matrix(xmin, xmax, ymin, ymax, pixel_density):
    re = np.linspace(xmin, xmax, int((xmax - xmin) * pixel_density))
    im = np.linspace(ymin, ymax, int((ymax - ymin) * pixel_density))
    return re[np.newaxis, :] + im[:, np.newaxis] * 1j

[38] def is_stable(c, num_iterations):
    z = 0
    for _ in range(num_iterations):
        z = z**17 + c
    return abs(z) <= 2

[39] def get_members(c, num_iterations):
    mask = is_stable(c, num_iterations)
    return c[mask]

[40] if __name__ == "__main__":
    c = complex_matrix(-2, 2, -2, 2, pixel_density=512)
    members = get_members(c, num_iterations=25)

    plt.scatter(members.real, members.imag, color="black", marker="s", s=1)
    plt.gca().set_aspect("equal")
    plt.axis("on")
    plt.tight_layout()
    plt.show()

```

Figure 2: Python implementation of the escape-time algorithm for generating generalized Mandelbrot sets. The script utilizes NumPy for vectorization and calculates membership based on the divergence of iterates within a defined bailout radius.

6 Numerical Area Estimates

In this section, I study the family

$$f(z) = z^j + c,$$

where $j \geq 3$, using the escape-time algorithm described above. For each value of j , I compute a numerical approximation of the area of the corresponding multibrot set M_j .

To make these approximations more reliable, I run the algorithm using different values of the iteration limit. Increasing the number of iterations allows the algorithm more time to detect points whose orbits eventually escape, while increasing the pixel density gives a finer approximation of the boundary and reduces sampling error.

The resulting data are summarized in Table 1 as ordered pairs of the form (j, area) . I then plot these values and fit an empirical curve to describe how the area changes as a function of j .

Table 1: Area approximations for generalized Mandelbrot sets of different degrees j , computed using the escape-time algorithm.

Degree j	Area
3	1.8
4	1.98
5	2.115
6	2.2175
7	2.2972
8	2.3625
9	2.4169
10	2.4628
11	2.5021
12	2.5364
13	2.5674
14	2.5942
15	2.6184
16	2.6410
17	2.6605
18	2.6793
19	2.6960
20	2.7115
50	2.9190
80	2.9851
200	3.0646

6.1 Numerical Convergence and Error Analysis

The numerical estimation of the Lebesgue measure

$$\text{Area}(M_j) = \iint_{\mathbb{C}} \chi_{M_j}(c) dA$$

is subject to two independent sources of approximation error. The first arises from replacing the infinite escape-time criterion by a finite iteration limit, while the second results from evaluating the resulting finite-iteration set on a discrete Cartesian sampling grid.

6.1.1 Finite-Iteration Approximation

For a fixed polynomial degree $j \geq 2$, define the finite-iteration approximation

$$M_{j,K} = \{c \in \mathbb{C} : |z_n(c)| \leq 2 \text{ for every } 1 \leq n \leq K\},$$

where

$$z_0(c) = 0, \quad z_{n+1}(c) = z_n(c)^j + c.$$

Since a parameter that escapes by iteration K must also escape for every larger iteration limit,

$$M_{j,K+1} \subseteq M_{j,K},$$

and

$$M_j = \bigcap_{K=1}^{\infty} M_{j,K}.$$

Each $M_{j,K}$ is compact, and the sequence $\{M_{j,K}\}_{K=1}^{\infty}$ forms a nested decreasing family of sets. By continuity of Lebesgue measure from above,

$$\lim_{K \rightarrow \infty} \text{Area}(M_{j,K}) = \text{Area}(M_j),$$

with

$$\text{Area}(M_{j,K+1}) \leq \text{Area}(M_{j,K}) \quad \text{for all } K.$$

Consequently, every finite-iteration approximation provides an upper bound for the true area of the corresponding Multibrot set.

6.1.2 Empirical Convergence with Respect to the Iteration Limit

To investigate the dependence of the computed areas on the iteration threshold, the escape-time algorithm was evaluated for

$$K = 25, 100, 500, 1000,$$

while holding the spatial sampling density fixed at a pixel density of 512.

Table 2: Empirical area estimates for generalized Multibrot sets computed using a fixed spatial sampling density and varying iteration limits.

Degree j	$K = 25$	$K = 100$	$K = 500$	$K = 1000$
3	1.8966	1.8161	1.7969	1.7954
4	2.0533	1.9971	1.9834	1.9819
5	2.1717	2.1271	2.1162	2.1149
6	2.2628	2.2628	2.2182	2.2169
7	2.3357	2.3054	2.2976	2.2967
8	2.3966	2.3699	2.3629	2.3621
9	2.4470	2.4239	2.4173	2.4165
10	2.4897	2.4687	2.4632	2.4624
11	2.5269	2.5077	2.5025	2.5017
12	2.5595	2.5419	2.5368	2.5361
13	2.5884	2.5723	2.5676	2.5671
14	2.6137	2.5988	2.5944	2.5939
15	2.6370	2.6227	2.6186	2.6181
16	2.6578	2.6451	2.6413	2.6407
17	2.6764	2.6647	2.6612	2.6608
18	2.6939	2.6829	2.6796	2.6790
19	2.7104	2.6994	2.6962	2.6958
20	2.7249	2.7145	2.7117	2.7113

As predicted by the nesting relation $M_{j,K+1} \subseteq M_{j,K}$, the computed areas are monotonically non-increasing across the tested iteration limits.

The largest corrections occur between $K = 25$ and $K = 100$. For example, for the degree-10 Multibrot set,

$$2.4897 \rightarrow 2.4687 \rightarrow 2.4632 \rightarrow 2.4624,$$

giving successive differences

$$0.0210, \quad 0.0055, \quad 0.0008.$$

A similar trend is observed throughout the tested range of degrees.

Table 3: Difference between the $K = 500$ and $K = 1000$ area estimates.

Degree j	$ A_{j,1000} - A_{j,500} $
3	0.0015
4	0.0015
5	0.0013
6	0.0013
7	0.0009
8	0.0008
9	0.0008
10	0.0008
11	0.0008
12	0.0007
13	0.0005
14	0.0005
15	0.0005
16	0.0006
17	0.0004
18	0.0006
19	0.0004
20	0.0004

Across all reported degrees, the difference between the $K = 500$ and $K = 1000$ estimates never exceeds approximately 1.5×10^{-3} . These results indicate that the computed areas become progressively less sensitive to further increases in the iteration threshold over the range considered in this study.

This observed stabilization should not, however, be interpreted as a rigorous error bound for $\text{Area}(M_j)$. A small difference between two finite-iteration approximations demonstrates only that those approximations are close to one another; it does not establish their distance from the limiting value $\text{Area}(M_j)$. Accordingly, the reported values should be interpreted as empirically stable numerical estimates rather than mathematically certified decimal approximations.

6.1.3 Spatial Discretization

A second source of approximation arises from evaluating $\chi_{M_{j,K}}$ on a finite Cartesian sampling grid. Because the boundary of a Multibrot set is highly irregular, grid cells intersecting the boundary may contain both escaping and non-escaping parameters, introducing additional discretization error whose magnitude depends on the sampling density.

In the present study, the spatial sampling density was held fixed while the iteration threshold was varied. Consequently, the convergence study above isolates the dependence of the computed areas on the iteration limit alone. A separate grid-refinement study would be required to quantify the contribution

of spatial discretization error.

Shishikura [18] proved that the boundary of the classical Mandelbrot set has Hausdorff dimension two. Although this result does not determine whether the boundary possesses positive two-dimensional Lebesgue measure, it illustrates why rigorous discretization error estimates for pixel-based area computations remain mathematically subtle.

For this reason, the empirical computations presented here are complemented in Section 9 by independent analytic upper bounds obtained from Gronwall’s Area Theorem. Together, these numerical and theoretical approaches provide mutually reinforcing evidence for the reported Multibrot area estimates.

6.2 Empirical Observations

Empirical Observation 1 *As j increases, the Multibrot sets become increasingly well approximated by the unit disk.*

Representative Multibrot sets for several values of j are shown in Figure 3, illustrating the increasingly circular geometry and smoother boundary as the degree increases. For smaller values of j , the boundary exhibits a pronounced gear-like structure. As j increases, the curvature becomes progressively more uniform, visually approaching that of the unit circle.

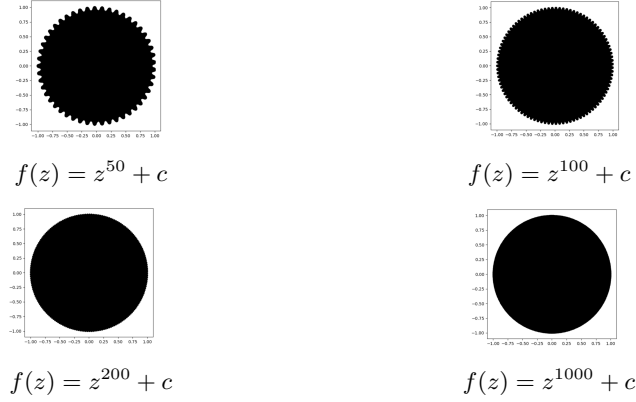


Figure 3: Representative Multibrot sets for $f(z) = z^j + c$. As the degree increases, the sets become increasingly circular and their boundaries progressively smoother. Images were generated using Python’s NumPy and Matplotlib libraries on a 1000×1000 grid with an escape-time algorithm using 200 iterations.

Empirical Observation 2 *As j increases, the effective radius of the Multibrot set decreases.*

The escape criterion established earlier shows that every Multibrot set is contained in a disk of radius 2. However, the numerical experiments in Figure 3 indicate that, for sufficiently large values of j , the sets become increasingly well approximated by the unit disk. Consequently, the effective radius appears to decrease toward 1 as j increases.

Empirical Observation 3 *The computed area $C(j)$ appears to approach π as $j \rightarrow \infty$.*

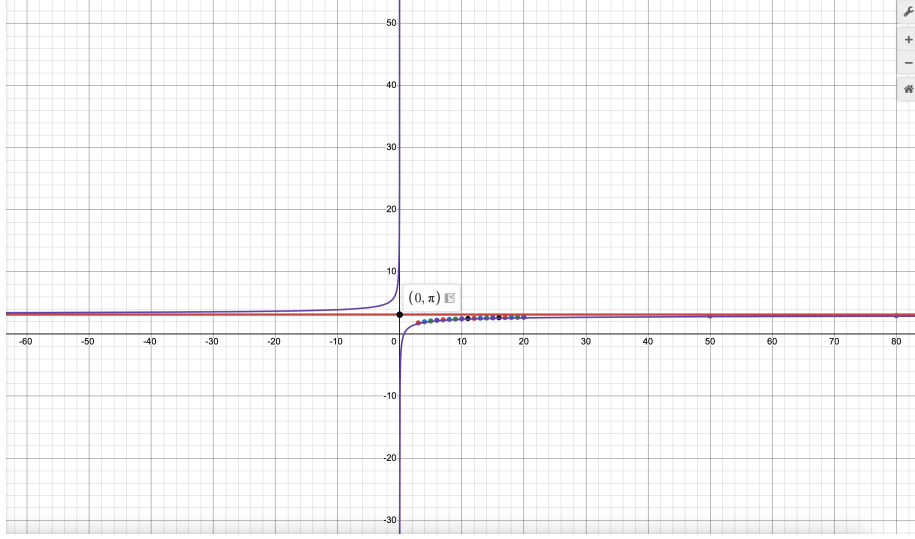


Figure 4: Computed area estimates of the Multibrot set for increasing polynomial degree j . Areas were estimated using escape-time simulations on a 1000×1000 grid and averaged across multiple iteration depths (full computational details are available in the GitHub repository). The data exhibit clear asymptotic convergence toward π , the area of the unit disk.

Figure 4 provides quantitative evidence supporting the geometric behavior observed in Figure 3. As the Multibrot sets become increasingly well approximated by the unit disk, their computed areas approach the area of the unit disk, namely π . The numerical data therefore support the asymptotic behavior suggested by the visual convergence of the sets.

Conjecture *The relationship between the polynomial degree j and the corresponding area approximation $C(j)$ follows a simple asymptotic trend.*

The numerical data are well approximated by the empirical model

$$C(j) = \pi - \frac{2.6057}{j^{0.5264}}.$$

This curve was obtained by fitting the computed area values shown in Figure 4. It captures the observed convergence of $C(j)$ toward π over the computed range of degrees and provides a simple quantitative model for the asymptotic behavior of the Multibrot set area. Although purely empirical, the fitted curve closely matches the numerical data and is consistent with the geometric observations presented above.

7 Symmetries

I now formally investigate the reflectional and rotational symmetries inherent to generalized Mandelbrot sets. These symmetries are governed by the dihedral group D_{j-1} , the symmetry group of a regular $(j-1)$ -gon. In particular, the generalized Multibrot sets exhibit both $(j-1)$ -fold rotational symmetry and $(j-1)$ reflection symmetries. Figures 5 for degrees $j \in \{4, 5, 6\}$ provide visual illustrations of these geometric properties.

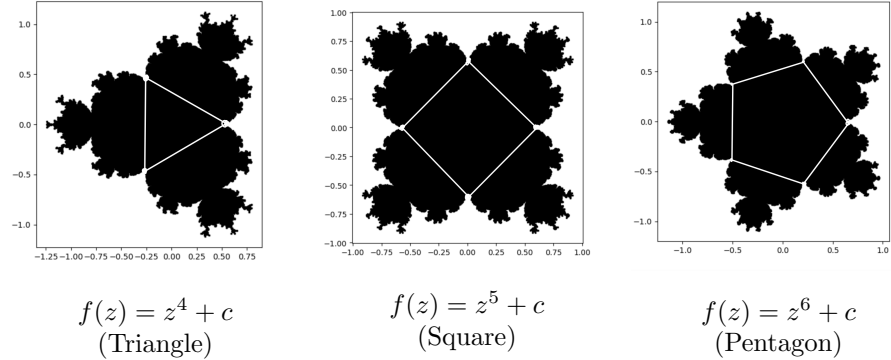


Figure 5: Roots of unity structures for $f(z) = z^j + c$ across $j = 4, 5, 6$. Regular $(j-1)$ -gons formed by roots of unity placed on the complex unit circle naturally bound the central core geometry of each generalized set.

More precisely, for a generalized set of degree j , the vertices of the inscribed polygon correspond to scaled $(j-1)$ -st roots of unity. Introducing a uniform scaling factor $k > 0$, constructing the explicit lines of symmetry reduces to evaluating the perpendicular bisectors originating from these dilated vertices through the origin.

Figure 6 illustrates the resulting symmetry frameworks and coordinate conversions for $j \in \{4, 5, 6\}$.

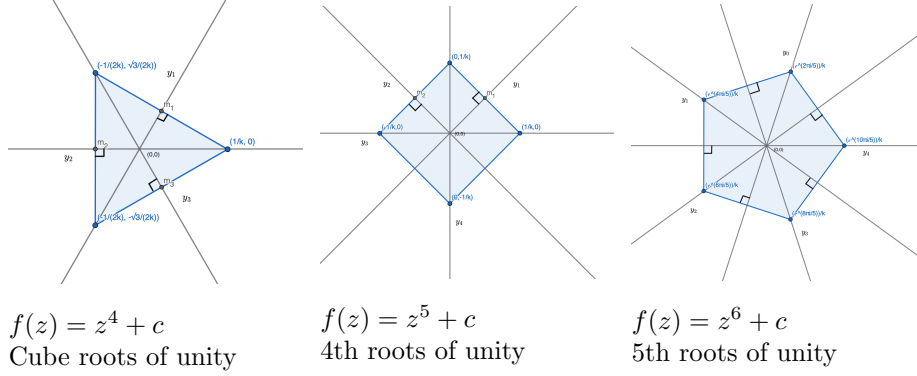


Figure 6: Symmetry line derivations for generalized Mandelbrot sets. Symmetry axes were constructed by converting the roots of unity $z^n = 1$ ($n = 3, 4, 5$) into Cartesian coordinates and calculating their perpendicular bisectors through the origin.

To demonstrate this explicit line derivation, consider $f(z) = z^4 + c$. The scaled roots of unity satisfy $(kz)^3 = 1 = e^{2\pi i}$, yielding:

$$\begin{aligned}
kz_0 = 1 &\implies z_0 = \left(\frac{1}{k}, 0\right) \\
kz_1 = e^{2\pi i/3} = -\frac{1}{2} + \frac{\sqrt{3}}{2}i &\implies z_1 = \left(-\frac{1}{2k}, \frac{\sqrt{3}}{2k}\right) \\
kz_2 = e^{4\pi i/3} = -\frac{1}{2} - \frac{\sqrt{3}}{2}i &\implies z_2 = \left(-\frac{1}{2k}, -\frac{\sqrt{3}}{2k}\right)
\end{aligned}$$

Evaluating the lines connecting each vertex through the origin $(0, 0)$ via slope-intercept form gives the three principal axes of reflection:

$$y_1 = \sqrt{3}x, \quad y_2 = 0, \quad y_3 = -\sqrt{3}x.$$

7.1 Formal Proof of Invariance

To rigorously prove reflectional and rotational invariance across generalized degrees, we define two transformation operators on $\mathbb{C} \cong \mathbb{R}^2$:

1. **Reflection Operator** $R_m(x, y)$: Reflection across a line through the origin with slope m :

$$R_m(x, y) = \left(\frac{(1 - m^2)x + 2my}{1 + m^2}, \frac{(m^2 - 1)y + 2mx}{1 + m^2} \right).$$

2. **Rotation Operator $r_k(c)$:** Rotation by discrete multiples of $\frac{2\pi}{j-1}$:

$$r_k(c) = c \cdot e^{\frac{2\pi i k}{j-1}}, \quad k \in \{0, 1, \dots, j-2\}.$$

The following technical lemmas establish the required algebraic preservation properties.

Lemma 7.1 *Let $T \in \{R_m, r_k\}$ be any reflection or rotation operator corresponding to degree $j \in \{4, 5, 6\}$. Then T preserves modulus:*

$$|T(z)| = |z| \quad \forall z \in \mathbb{C}.$$

Lemma 7.2 *For each transformation T specified in Lemma 7.1, T commutes with the polynomial exponentiation step:*

$$T(z^j) = (T(z))^j \quad \forall z \in \mathbb{C}.$$

Lemma 7.3 *Let $z_{n,c}$ denote the n -th iterate of $f_c(z) = z^j + c$ starting from $z_{0,c} = 0$. For any transformation T satisfying Lemmas 7.1 and 7.2:*

$$z_{n,T(c)} = T(z_{n,c}) \quad \forall n \geq 0.$$

We proceed by induction on the iteration depth n .

- **Base Case ($n = 0$):** By definition, $z_{0,T(c)} = 0$. Since $T(0) = 0$, $z_{0,T(c)} = T(z_{0,c})$ holds trivially.
- **Inductive Step:** Assume $z_{N,T(c)} = T(z_{N,c})$ holds for some $N \geq 0$. Applying the recurrence relation and Lemma 7.2:

$$\begin{aligned} z_{N+1,T(c)} &= (z_{N,T(c)})^j + T(c) \\ &= (T(z_{N,c}))^j + T(c) \quad (\text{by inductive hypothesis}) \\ &= T(z_{N,c}^j) + T(c) \quad (\text{by Lemma 7.2}) \\ &= T(z_{N,c}^j + c) \quad (\text{by linearity of } T) \\ &= T(z_{N+1,c}). \end{aligned}$$

By induction, $z_{n,T(c)} = T(z_{n,c})$ for all $n \in \mathbb{N}_0$.

7.2 Proof of Lemma Properties for $j = 6$

To demonstrate the concrete algebraic verification of Lemmas 7.1 and 7.2, we present the proof for the most computationally intensive reflection case: $j = 6$ across the line $y = \tan\left(\frac{2\pi}{5}\right)x$. The full derivations for all remaining reflectional cases ($j = 4, 5$) and rotational symmetry cases are provided in the Appendix and archived in the project GitHub repository.

[Proof of Lemma 7.1 for $j = 6$] Let $z = a + bi \cong (a, b)$ and let $m = \tan\left(\frac{2\pi}{5}\right)$. Using $1 + m^2 = \sec^2\left(\frac{2\pi}{5}\right)$, the reflection operator simplifies to:

$$R(a, b) = \left(\frac{(1 - m^2)a + 2mb}{\sec^2\left(\frac{2\pi}{5}\right)}, \frac{(m^2 - 1)b + 2ma}{\sec^2\left(\frac{2\pi}{5}\right)} \right).$$

Evaluating $|R(z)|^2$:

$$\begin{aligned} |R(a, b)|^2 &= \frac{[(1 - m^2)a + 2mb]^2 + [(m^2 - 1)b + 2ma]^2}{\sec^4\left(\frac{2\pi}{5}\right)} \\ &= \frac{(1 - m^2)^2 a^2 + 4m(1 - m^2)ab + 4m^2 b^2 + (m^2 - 1)^2 b^2 + 4m(m^2 - 1)ab + 4m^2 a^2}{(1 + m^2)^2} \\ &= \frac{(1 + 2m^2 + m^4)a^2 + (1 + 2m^2 + m^4)b^2}{(1 + m^2)^2} \\ &= \frac{(1 + m^2)^2(a^2 + b^2)}{(1 + m^2)^2} = a^2 + b^2 = |z|^2. \end{aligned}$$

Taking the square root yields $|R(z)| = |z|$.

[Proof of Lemma 7.2 for $j = 6$] Expanding $z^6 = (a + bi)^6$ via the Binomial Theorem yields:

$$z^6 = (a^6 - 15a^4b^2 + 15a^2b^4 - b^6) + i(6a^5b - 20a^3b^3 + 6ab^5).$$

Applying the reflection operator directly to z^6 gives:

$$\begin{aligned} R(z^6) &= \frac{(1 - m^2)(a^6 - 15a^4b^2 + 15a^2b^4 - b^6) + 2m(6a^5b - 20a^3b^3 + 6ab^5)}{\sec^2\left(\frac{2\pi}{5}\right)} \\ &\quad + i \left[\frac{(m^2 - 1)(6a^5b - 20a^3b^3 + 6ab^5) + 2m(a^6 - 15a^4b^2 + 15a^2b^4 - b^6)}{\sec^2\left(\frac{2\pi}{5}\right)} \right]. \end{aligned}$$

Conversely, expanding $(R(z))^6$ symbolically confirms exact agreement term-by-term:

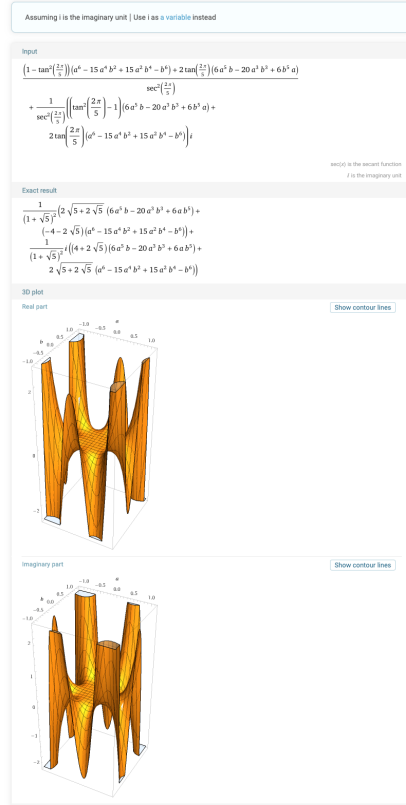
$$(R(z))^6 = R(z^6).$$

While the symbolic expansion involves extensive algebra, Figure 7 provides numerical and visual verification of this identity across the complex domain.

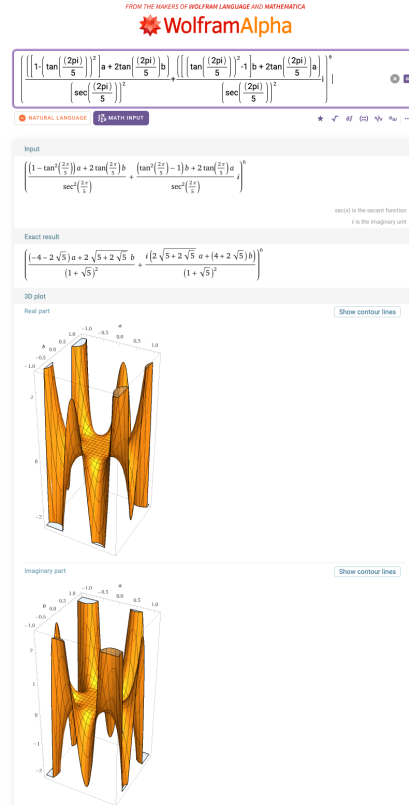
7.3 Main Symmetry Theorem

Equipped with Lemmas 7.1–7.3, we state and prove the primary symmetry theorem governing generalized Mandelbrot sets.

Theorem 7.4 (Symmetry Invariance) *Let $\mathcal{M}_j$ denote the generalized Mandelbrot set of degree j , and let $T \in \{R_m, r_k\}$ be any symmetry transformation satisfying Lemmas 7.1 and 7.2. If $c \in \mathcal{M}_j$, then $T(c) \in \mathcal{M}_j$.*



$$R(z^6)$$



$$(R(z))^6$$

Figure 7: Surface plot comparison of $R(z^6)$ vs $(R(z))^6$. The complete alignment of real and imaginary surface manifolds confirms the operational commutation $R(z^6) = (R(z))^6$.

Suppose $c \in \mathcal{M}_j$. By definition, the sequence of iterates $z_{n,c}$ remains bounded; that is, there exists some $B > 0$ such that $|z_{n,c}| < B$ for all $n \in \mathbb{N}_0$.

By Lemma 7.3, $z_{n,T(c)} = T(z_{n,c})$ for all n . Applying Lemma 7.1:

$$|z_{n,T(c)}| = |T(z_{n,c})| = |z_{n,c}| < B \quad \forall n \in \mathbb{N}_0.$$

Thus, the sequence of iterates $z_{n,T(c)}$ remains bounded by B . Consequently, $T(c) \in \mathcal{M}_j$, establishing that $\mathcal{M}_j$ is invariant under T .

8 Morphological Evolution and Asymptotic Circularity

A striking feature of the generalized family $f(z) = z^j + c$ is the structural evolution of the main body as the polynomial degree j increases. For any finite degree $j \geq 3$, the set exhibits $j - 1$ distinct boundary lobes or "petals." These feature components directly reflect the discrete reflectional and rotational symmetries established in Section 7.

Crucially, as j increases, the angular spacing between adjacent symmetry axes ($\frac{2\pi}{j-1}$) decreases monotonically. This proliferation of lobes induces a visual and geometric smoothing effect: the discrete $(j - 1)$ -gonal boundary structures become progressively finer and densely packed around the complex unit circle $\mathbb{T} = \{c \in \mathbb{C} : |c| = 1\}$.

In the asymptotic limit $j \rightarrow \infty$, the discrete rotational invariance transitions into continuous S^1 rotational symmetry, causing the macroscopic boundary of $\mathcal{M}_j$ to converge toward the unit disk $D_1(0)$. This structural compression provides the geometric intuition underlying the area convergence $A(\mathcal{M}_j) \rightarrow \pi$ demonstrated in Section 6.

Figure 8 illustrates these structural features across degrees $j \in \{4, 6, 7, 10\}$.

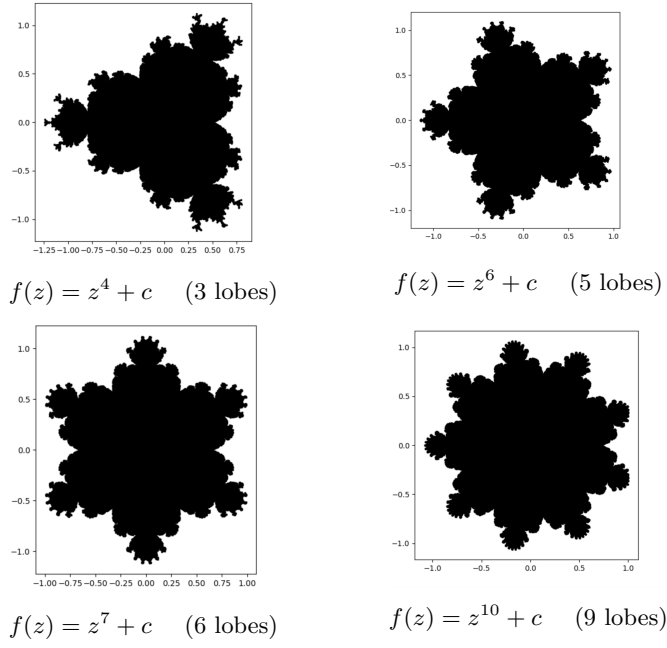


Figure 8: Morphological petal structures across increasing polynomial degrees j . Each set of degree j exhibits $(j-1)$ -fold rotational symmetry. As j increases, the primary lobes multiply, sharpen, and uniformly compress toward the complex unit circle $\mathbb{T}$, driving the set's macroscopic geometry toward circularity.

9 Validation Using Laurent Coefficients and Gronwall's Formula

The area estimates presented in Section 6 were derived via escape-time grid sampling. To rigorously evaluate the stability and accuracy of these numerical computations, we introduce an independent, conformal-analytic verification framework based on the classical area theorem of Gronwall.

Gronwall's area formula directly computes the area enclosed by the image of a univalent conformal map using the coefficients of its Laurent series expansion. By constructing the exterior Riemann map for generalized Multibrot sets, we obtain explicit analytic upper bounds on the enclosed planar area.

9.1 The Riemann Map and Inverse Böttcher Expansion

By the Riemann Mapping Theorem, there exists a unique conformal isomorphism $\psi : \{z \in \mathbb{C} : |z| > 1\} \rightarrow \mathbb{C} \setminus \mathcal{M}_j$ mapping the exterior of the closed unit disk onto the exterior of the degree- j Multibrot set $\mathcal{M}_j$, subject to the normalization at infinity $\psi(z) = z + \mathcal{O}(1)$.

Near infinity, $\psi(z)$ admits a Laurent series expansion of the form:

$$\psi(z) = z + b_0 + \sum_{m=1}^{\infty} \frac{b_m}{z^m}, \quad b_m \in \mathbb{C}. \quad (1)$$

The coefficients b_m quantify the geometric distortion of the boundary $\partial\mathcal{M}_j$ relative to the unit circle and serve as the direct inputs for Gronwall's area calculation.

9.2 Adapting and Analytically Verifying Laurent Coefficient Computations

To construct the Laurent coefficients for generalized Multibrot sets, I modified an algorithm originally developed for the classical quadratic Mandelbrot set (the Jungreis algorithm for $f(z) = z^2 + c$), for arbitrary degree.

To validate the modified implementation, I analytically compute the first several terms of the Laurent expansion for $f(z) = z^4 + c$. The exterior Riemann map satisfies the functional equation induced by the Böttcher coordinate near infinity:

$$\psi(z^4) = (\psi(z))^4 + c. \quad (2)$$

Expanding both sides of Eq. (2) in powers of z :

$$\begin{aligned}\psi(z^4) &= z^4 + b_0 + \frac{b_1}{z^4} + \frac{b_2}{z^8} + \frac{b_3}{z^{12}} + \dots \\ (\psi(z))^4 &= \left(z + b_0 + \frac{b_1}{z} + \frac{b_2}{z^2} + \frac{b_3}{z^3} + \frac{b_4}{z^4} + \frac{b_5}{z^5} + \dots \right)^4 + c \\ &= z^4 + 4b_0z^3 + 4b_1z^2 + 4b_2z + (4b_3 + c) + \frac{4b_4}{z} + \frac{4b_5}{z^2} + \mathcal{O}(z^{-3}).\end{aligned}$$

Equating corresponding powers of z yields the decoupled linear system:

$$\begin{aligned}z^3 : \quad & 4b_0 = 0 \implies b_0 = 0 \\ z^2 : \quad & 4b_1 = 0 \implies b_1 = 0 \\ z^1 : \quad & 4b_2 = 0 \implies b_2 = 0 \\ z^0 : \quad & 4b_3 + c = b_0 = 0 \implies b_3 = -\frac{1}{4}c \\ z^{-1} : \quad & 4b_4 = 0 \implies b_4 = 0 \\ z^{-2} : \quad & 4b_5 = 0 \implies b_5 = 0\end{aligned}$$

For the representative parameter $c = 1$, the leading non-zero coefficient evaluates to $b_3 = -\frac{1}{4} = -0.25$. As shown in Figure 9, the automated output of our modified numerical algorithm matches this analytic result precisely, verifying the algorithmic modification.

```

EXPLORER  PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
LAURENT SERIES MULTIBROT
  calculate_area.py
  generate_coefficients.py

/usr/bin/python3 "/Users/rohansenapati/Downloads/laurent series multibrot/generate_coefficients.py"
rohansenapati@MacBook-JS406: laurent series multibrot % /usr/bin/python3 "/Users/rohansenapati/Downloads/laurent series multibrot/generate_coefficients.py"
Degree d = 4 ---
Coefficients:
b_1 = 0.0
b_2 = 0.0
b_3 = -0.25
b_4 = 0.0
b_5 = 0.0
b_6 = 0.046875
b_7 = 0.0
b_8 = 0.0
b_9 = -0.005859375
b_10 = 0.0
b_11 = 0.0
b_12 = 0.00018318546875
b_13 = 0.0
b_14 = 0.0
b_15 = -0.06238555980283125
b_16 = 0.0
b_17 = 0.0
b_18 = 0.023418328281982422
b_19 = 0.0
b_20 = 0.0
b_21 = -0.004393279552459717
b_22 = 0.0
b_23 = 0.0
b_24 = 0.00018391292542219162
b_25 = 0.0
b_26 = 0.0
b_27 = 0.0001428387732567191
b_28 = 0.0
b_29 = 0.0
b_30 = -0.0018173285737373792
b_31 = 0.0
b_32 = 0.0
b_33 = 0.0013449714415401184
b_34 = 0.0
b_35 = 0.0
b_36 = -0.0007536973323851726
b_37 = 0.0
b_38 = 0.0
b_39 = 0.00024269123023756833
b_40 = 0.0
b_41 = 0.0
b_42 = -4.1545481227915946e-05
b_43 = 0.0
b_44 = 0.0
b_45 = 2.9364884239908844e-05
b_46 = 0.0
b_47 = 0.0
b_48 = 2.9458856431118276e-05
b_49 = 0.0
b_50 = 0.0

Area: 2.2941246874837376
rohansenapati@MacBook-JS406: laurent series multibrot %

```

Figure 9: Numerical output of the modified Laurent coefficient implementation for $f(z) = z^4 + c$. The leading non-zero coefficient $b_3 = -0.25$ agrees exactly with the symbolic derivation from the inverse Böttcher functional equation $\psi(z^4) = (\psi(z))^4 + c$.

The observed sparsity of the Laurent coefficients in Figure 9 is consistent with analogous phenomena previously reported for the classical Mandelbrot set, where many coefficients vanish identically. My computations suggest that similar structural behavior persists for generalized Multibrot sets.

As a secondary qualitative check, I reconstructed the boundary contours of $\mathcal{M}_j$ directly from truncated Laurent series maps. Figure 10 shows the reconstructed exterior mappings for $j \in \{3, 4, 6\}$, faithfully reproducing the expected $(j - 1)$ -fold rotational symmetry.

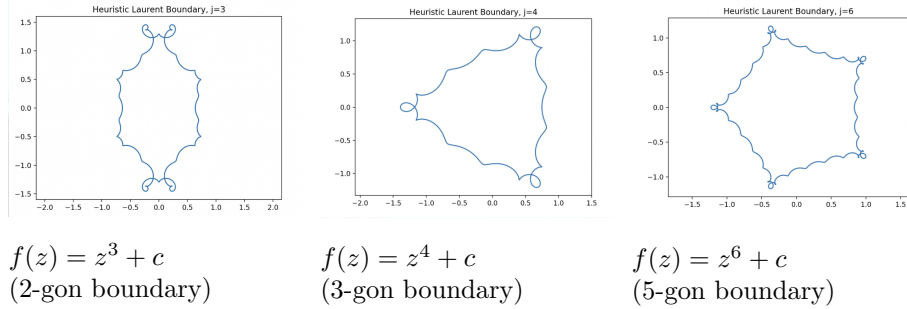


Figure 10: Boundary approximations generated from truncated Laurent series. Reconstructions for degrees $j = 3, 4$, and 6 confirm that the generated coefficients accurately capture the geometric $(j - 1)$ -fold symmetry of the underlying Multibrot sets.

9.3 Gronwall's Area Formula and Upper Bounds

Given the Laurent coefficients $(b_1, b_2, \dots)$ of the exterior Riemann map $\psi(z)$, Gronwall's Area Theorem asserts that the total area $A(\mathcal{M}_j)$ enclosed by the boundary $\partial\mathcal{M}_j$ is given by:

$$A(\mathcal{M}_j) = \pi \left(1 - \sum_{m=1}^{\infty} m |b_m|^2 \right). \quad (3)$$

In practice, evaluating Eq. (3) computationally requires a finite truncation at degree M :

$$A_M(\mathcal{M}_j) = \pi \left(1 - \sum_{m=1}^M m |b_m|^2 \right). \quad (4)$$

Because every term in the sum $m|b_m|^2$ is strictly non-negative, any finite truncation $A_M(\mathcal{M}_j)$ overestimates the true enclosed area, providing a strict, rigorous upper bound:

$$A(\mathcal{M}_j) \leq A_M(\mathcal{M}_j) \quad \forall M \in \mathbb{N}.$$

I computed truncated Laurent series up to order $M = 50$ and evaluated Gronwall upper bounds for Multibrot degrees $j \in \{3, 4, \dots, 20\}$.

9.4 Comparison with Grid-Sampling Estimates

Figure 11 overlays the conformal-analytic Gronwall upper bounds against the computational grid-sampling area estimates derived in Section 6.

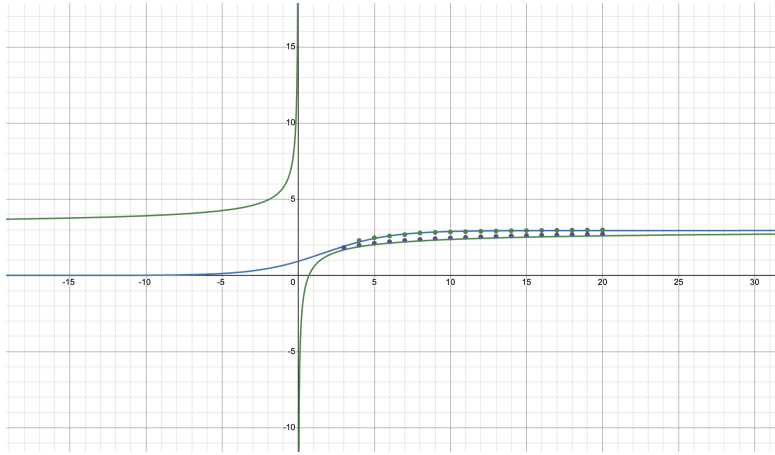


Figure 11: Comparison between conformal Gronwall area upper bounds and grid-sampling area estimates. A logistic regression fitted to the Gronwall bounds (blue) strictly encapsulates the empirical area curve derived from escape-time grid sampling (green) across all degrees $j \in [3, 20]$. The upper bound envelope approaches π , providing conformal-analytic support for the asymptotic disk-limit behavior observed in the numerical simulations.

Across all evaluated degrees $j \in \{3, \dots, 20\}$, the grid-sampling estimates lie strictly below the independent Gronwall upper bounds. The agreement between these two distinct methodologies—one measure-theoretic/computational, the other complex-analytic—validates the numerical simulation pipeline and confirms that the asymptotic convergence $A(\mathcal{M}_j) \rightarrow \pi$ reflects genuine underlying complex geometry.

10 Conclusion

This paper studied the geometry of multibrot sets from complementary computational and conformal perspectives. Computationally, large-scale escape-time simulations suggest a smooth progression of multibrot area toward the area of the unit disk as the polynomial degree increases, leading to an empirical model describing this convergence. Geometrically, explicit elementary derivations of reflectional and rotational symmetries for representative multibrot sets, together with the emergence of increasingly dense $(j-1)$ -gon structures, provide an intuitive explanation for the observed approach toward circularity. From the conformal perspective, the Laurent-coefficient framework together with Gronwall’s area formula provides an independent analytic validation of the numerical area computations, linking empirical observations with classical techniques from complex analysis.

Beyond the computational experiments, this work demonstrates how numerical and conformal methods can complement one another in the study of polynomial parameter spaces. While the empirical area model is consistent with the computed data and the conformal area bounds obtained from Laurent coefficients, a theoretical derivation of the proposed model remains an open problem. Likewise, although explicit symmetry derivations are presented for representative cases, extending these arguments and further investigating the relationship between symmetry, conformal geometry, and the global geometry of multibrot sets remain natural directions for future work.

Future work may also investigate machine learning and other computational techniques to explore higher-degree multibrot sets, test additional conjectures, and identify new geometric patterns. More broadly, the conformal methods employed in this work demonstrate how classical tools from complex analysis can provide valuable independent validation of computational investigations, offering a unified perspective on how multibrot geometry evolves as the polynomial degree increases.

11 Acknowledgments

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Appendix

The following materials provide the formal algebraic foundations and high-resolution datasets supporting the conjectures in this study. To maintain a focused discussion on the geometric implications, the comprehensive proofs and raw simulation data are hosted in the following GitHub repository :

https://github.com/frostghost555/rohan_senapati-mandelbrot-appendix.git

This repository includes:

- **Formal Proofs:** Step-by-step proofs for the rotational and reflectional symmetry lemmas, along with specific cases for the escape-criterion proof and symmetry line derivations.
- **Area Convergence Data:** The empirical pixel-density values used to derive the area-degree curve $C(j)$.

This repository ensures that mathematically advanced readers and reviewers can validate the key steps in the logical reasoning of my proofs, while keeping the main manuscript focused and readable.